(2)

2 estados

Sempre serão concertadas até 2 máquinas




$p_{32} = (1-p)^3$

$p_{33} = 1 - p_{32}$

$p_{22} = (1-p)^2$

$p_{23} = 1 - p_{22}$

$X_0 = [1 \quad 0]$

$X_2 = [0{,}992 \quad 0{,}008]$

$\downarrow$

$X_2 = X_0 \cdot p^2$

Regime Estacionário

$E_2 \cdot p_{23} = E_3 \cdot p_{32}$

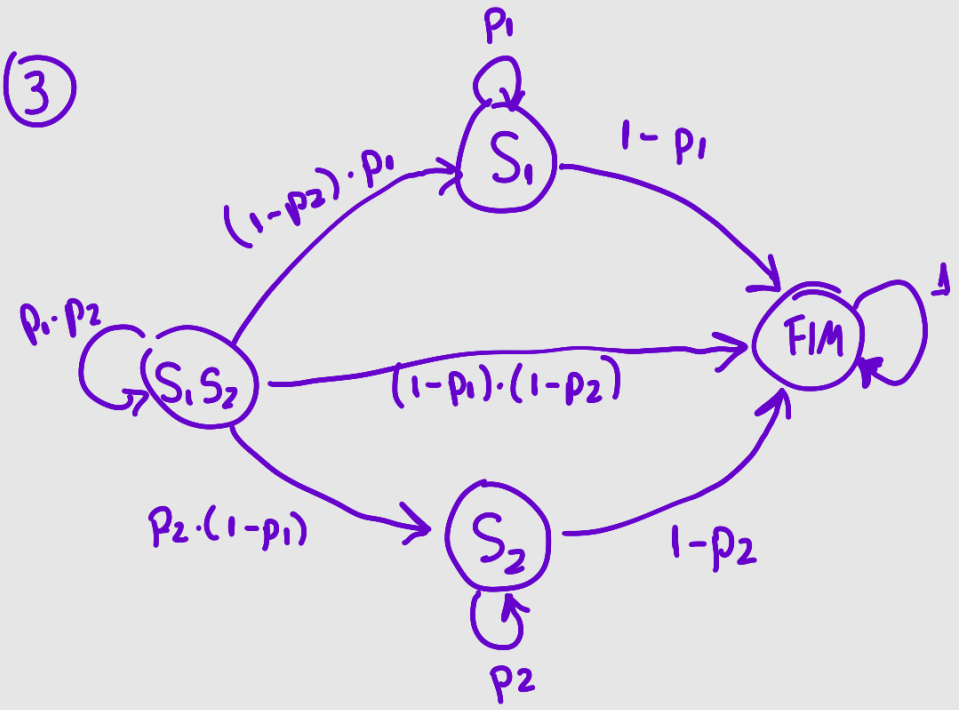
$E_2 + E_3 = 1$

$0,96 \cdot E_2 = 0,008 \cdot E_3$

$E_3 = 0,992$

---

③




②

$$\begin{array}{r} 0,2 \\ \times 0,2 \\ \hline 0,04 \end{array}$$

$$\begin{array}{r} \times \quad 0,2 \\ \hline 0,008 \end{array}$$




$p_{32} = (1-p)^3 = 0,008$

$$P = \begin{array}{c} \\ 3 \\ 2 \end{array} \begin{array}{c} \begin{array}{cc} 3 & 2 \end{array} \\ \begin{bmatrix} p_{33} & p_{32} \\ p_{23} & p_{22} \end{bmatrix} \end{array}$$

$p_{33} = 1 - p_{32} = 0{,}992$

$p_{22} = (1-p)^2 = 0{,}04$

$p_{23} = 1 - p_{22} = 0{,}96$

$$= \begin{bmatrix} 0{,}992 & 0{,}008 \\ 0{,}96 & 0{,}04 \end{bmatrix}$$

$0{,}00768$

$$\begin{bmatrix} p_{33} & p_{32} \\ p_{23} & p_{22} \end{bmatrix} \cdot \begin{bmatrix} p_{33} & p_{32} \\ p_{23} & p_{22} \end{bmatrix}$$

$$= \begin{bmatrix} p_{33} \cdot p_{33} + p_{32} \cdot p_{23} & p_{33} \cdot p_{32} + p_{32} \cdot p_{22} \\ p_{23} \cdot p_{33} + p_{22} \cdot p_{23} & p_{23} \cdot p_{32} + p_{22} \cdot p_{22} \end{bmatrix}$$

$$p^2 = \begin{bmatrix} 0{,}991744 & 0{,}008256 \\ 0{,}99072 & 0{,}00928 \end{bmatrix}$$

$$X_2 = X_0 \cdot p^2$$

$$\begin{bmatrix} 1 & 0 \end{bmatrix} \cdot \begin{bmatrix} 0{,}991744 & 0{,}008256 \\ 0{,}99072 & 0{,}0928 \end{bmatrix}$$

$X_2 = [0,991744 \quad 0,008256]$